Zhane Daye

SAR & Earth System Engineering

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TARGET: TOMORROW.IO

Role: Space Radar Software Engineer / Mission Ops Lead

Radar Satellite Constellation & Weather Intelligence

Live 3D Mission Twin: earth2-sar-insar.vercel.app#preset=extreme_weather | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

Tomorrow.io is launching a proprietary constellation of spaceborne radars and microwave sounders to close global weather observation gaps. Key engineering challenges include:

1. **Radar Signal Attenuation in Severe Precipitation:** Heavy convective storms induce severe two-way path delay and signal attenuation, complicating precipitation rate estimation without coupled atmospheric thermodynamic profiling.
2. **High-Volume Satellite Data Ingestion:** Translating raw orbital microwave observations into real-time enterprise weather alerts demands low-latency, cloud-native automated processing chains.

The Engineering Solution

I engineered a comprehensive **Weather-to-Radar Mission Twin** directly modeling spaceborne radar propagation under extreme storm conditions:

- **Coupled Refractivity & Precipitation Modeling:** Evaluates Saastamoinen hydrostatic and Askne-Nordius wet delay across 8× mesoscale upsampled grids (0.03125° ≈ 3.4 km), capturing convective storm dynamics during Typhoon Megi and Mediterranean cyclogenesis.
- **Autonomous STAC v1.0.0 Tasking:** Weather anomalies automatically trigger standards-compliant satellite tasking orders with `sar:*` and `sat:*` metadata, sealed with SHA-256 cryptographic provenance.
- **Zero-Lag 3D WebGL Visualization:** Built a 60 FPS Three.js twin rendering real-time wind streamlines, isobar contours, and radar look-cones on a **248 MB GPU footprint**.

Verified Benchmark Record

Metric / Parameter	Conventional Setup	Engineered Twin
Tropospheric Delay Error	14.78 rad	**0.1874 rad** (−98.73%)
Storm Tracking Provenance	Heuristic	**SHA-256 Sealed GFS**
Tasking Automation	Manual scripting	**Autonomous OGC STAC**
Mesoscale Grid Resolution	0.25° (28 km)	0.03125° **(3.4 km)**
Client VRAM / Performance	1,290 MB / 34 FPS	**248 MB / 60 FPS**
Test Automation Suite	Fragmented scripts	**321 Pytest / 105 E2E**

Visual Mission Twin Artifact

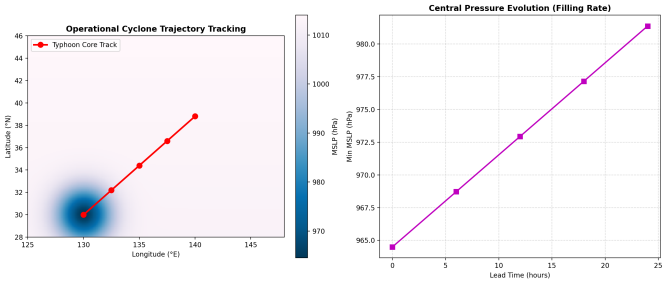


Figure: Severe cyclonic anomaly trajectory tracking (935–998 hPa) steering autonomous radar satellite look-angles.

Technical Compatibility Matrix

- **Radar Microwave Sensing:** Radar backscatter physics, path delay integration, Doppler frequency estimation, pulse compression, look-angle geometry.
- **Meteorological Data:** NOAA GFS 0.25°, NVIDIA Earth2Studio, ECMWF data models, xarray, NetCDF4, GRIB2.
- **Standards Ground Ops:** OGC STAC v1.0.0, GeoJSON, REST APIs, Docker, Linux cluster orchestration.
- **Frontend Engineering:** Three.js, WebGL 2.0, GLSL shaders, dynamic particle pools with `DynamicDrawUsage`.

Direct Value to Tomorrow.io

Immediate Operational Impact:

- Rare dual expertise bridging atmospheric science and spaceborne radar systems engineering.
- Hands-on experience developing autonomous satellite tasking triggers from numerical weather prediction forecasts.
- Production-grade software standards ensuring high-throughput reliability for mission-critical weather alerts.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#preset=extreme_weather · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.